ALBUKHARY INTERNATIONAL UNIVERSITY

BACHELOR IN COMPUTER SCIENCE (HONOURS)
BACHELOR IN DATA SCIENCE (HONOURS)

1. Comments in Python

Comments are lines in your code that Python ignores when running the program. They help explain what your code does or temporarily disable certain parts of the code.

Single-line comment: Start with a #

```
# This is a single-line comment
print("Hello, World!")
```

Inline comment: Add a comment at the end of a line

```
print("Hello, AI Class!")  # This prints a greeting
```

1.1 Multi-line Comments

Python doesn't have a dedicated multi-line comment syntax, but you can achieve this using:

Multiple # symbols:

```
# This is a
# multi-line comment
print("Hello, World!")
```

Triple quotes (""" or ''') — Python treats unassigned string literals as comments:

```
"""
This is a multi-line comment.
```

```
Useful for longer explanations.
"""
print("Hello, World!")
```

2. Variables in Python

Variables store data values. You create a variable by assigning a value to a name.

Basic variable creation:

```
name = "AI Class"
age = 10
```

Changing variable types:

```
x = 5        # x is an integer
x = "AI"     # x is now a string

print(x)
```

2.1 Variable Naming Rules

- Must start with a letter or underscore
- Can't start with a number
- Can only contain alphanumeric characters and underscores
- Case-sensitive (age, Age, and AGE are different)

Examples of **valid** variable names:

```
my_var = "AI"
myVar = 10
_myVar = 3.14
MYVAR = True
```

Invalid variable names:

```
2myvar = "AI"  # Can't start with a number
my-var = 10    # Hyphens are not allowed
my var = 5     # Spaces are not allowed
```

2. Variables in Python

2.2 Multi-word Variable Names

Camel Case: First word is lowercase, subsequent words start with a capital letter

```
myVariableName = "AI Class"
```

Pascal Case: Each word starts with a capital letter

```
MyVariableName = "AI Class"
```

Snake Case: Words are separated by underscores

```
my_variable_name = "AI Class"
```

2.3 Assigning Multiple Values

You can assign multiple variables at once:

```
x, y, z = "Apple", "Banana", "Cherry"
print(x)
print(y)
print(z)

Or assign the same value to multiple variables:

x = y = z = "Orange"
print(x, y, z)

x = "awesome"
print("Python is " + x)

x = "Python is "
y = "awesome"
z = x + y
print(z)
```

3. Python Data Types

Python has several built-in data types:

- Text: str
- Numbers: int, float, complex
- Sequence: list, tuple, range
- Mapping: dict
- Set: set, frozenset

- ## 2. Variables in Python
  - Binary: bytes, bytearray, memoryview

Example:

```
x = 5                # int
y = 3.14             # float
z = "Hello"          # str
is_python_fun = True # bool
```

4. Type Conversion

You can convert data types with these functions:

- *int() – Converts to integer*
- *float() – Converts to floating point number*
- *str() – Converts to string*

```
x = 5
y = 2.8
z = "3"

# Convert from int to float
a = float(x)

# Convert from float to int
b = int(y)

# Convert from string to int
c = int(z)

print(a, b, c)
```